Logical Pluralism and Topic-Neutralism

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The Fate of Logical Pluralism

Institute for Philosophical Research, UNAM,
Mexico City, 8–9 September 2026

Aims

- Explicate topic-specificism as a view about logic.
- Examine and assess several conceptions on how the laws of logic change.
- Propose an account in which the laws of logic change without any appeal to logic-specificism.
- Show that this account is compatible with a broadly classical conception of logic.

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What Is Logic?

Aristotle, *Analytica Priora*, c. 350 BC

We must first state what our inquiry is about and what its object is, saying that it is about demonstration and that its object is demonstrative science. (I 24a 10–12)

What Is Logic?

Petrus Hispanus, *Tractatus I*, written 1220s–1250s

Dialectic [i.e., Logic] is the art that provides the way to the first principles of all methods. Therefore, in the acquisition of the sciences, dialectic ought to come first. (De Dialectica, 4–5)

What Is Logic?

I. Kant, *Kritik der reinen Vernunft*, 2nd edition, 1787

It is not an enlargement but a disfigurement of the sciences when one allows their boundaries to run into one another. The boundary of logic, however, is thereby determined with complete precision: it is the science that expounds in detail and rigorously proves nothing but the formal rules of all thinking – whether that thinking be *a priori* or empirical, whatever its origin or object may be ... (Preface, pp. viii–ix)

What Is Logic?

Gilbert Ryle, *Dilemmas*, 1953/1964

Aristotle was investigating the logical powers of certain topic-neutral concepts, namely those of all, some and not. These are sometimes listed among what are nowadays called the 'logical constants'. (p. 115)

[Demarcation of Formal Logic from philosophy:] Formal Logic, it might be said, maps the inference-powers of the topic-neutral expressions or logical constants on which our arguments pivot; philosophy has to do with the topical or subject-matter concepts which provide the fat and the lean, but not the joints or the tendons of discourse. The philosopher examines such notions as *pleasure*, *colour*, *the future*, and *responsibility*, while the Formal Logician examines such notions as *all*, *some*, *not*, *if* and *or*. (p. 116)

What Is Logic?

Jan Woleński, 'The universality of logic', 2017

[T]hree understanding[s] of the universality property of logic might be discriminated:

(a) Logic is universal, if it is universally applicable;
(b) Logic is universal, if it is topic-neutral;
(c) Logic is universal, if its principles are universally true.

According to (a), every science as well as the commonsensical knowledge assume and apply logical rules, consciously or not. The view (b) denies that logic privileges any concrete subject. The point (c) claims that logical theorems are true in all circumstances, situations, states of the world, etc. (p. 23)

The Topic of Logic

The topic of logic

Logic is a discipline about correct reasoning or correct argumentation or correct deduction, among other possibilities. Hence, logic has a *topic* – in an intuitive sense of ‘topic’.

Warning

This is consistent with the idea that logic is topic-neutral. For the sense of the topic neutrality thesis is that the principles of reasoning, argumentation, or deduction are valid *regardless of the topic being reasoned, argued, or deducted about*.

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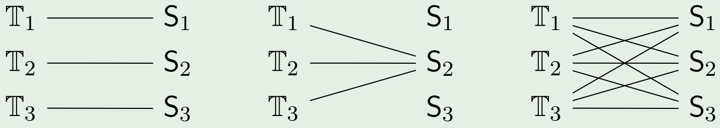
Views about logic

Views about logic

A view about logic is not just an inference system. However, it may be conceived as a collection of statements indicating which inference systems are correctly applicable to which topics.

Diagrammatic representations



Nihilism

For every topic, no logic system correctly applies to it.

Diagrammatic representations

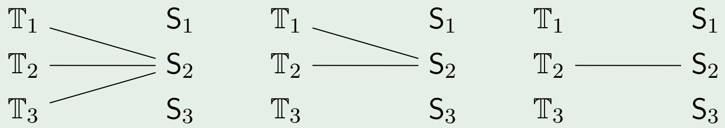
$\mathbb{T}$	S	$\mathbb{T}_1$, $\mathbb{T}_2$	S_1, S_2	$\mathbb{T}_1$, $\mathbb{T}_2$, $\mathbb{T}_3$	S

Monism

Monism

For every topic, no more than one inference system is correctly applicable to it.

Diagrammatic representations

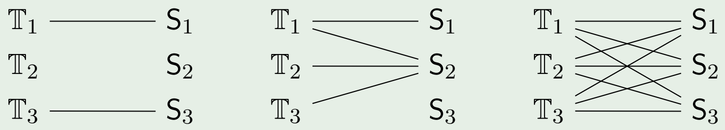


Pluralism

Pluralism

There are at least two inference systems applicable to some topic (probably distinct ones).

Diagrammatic representations

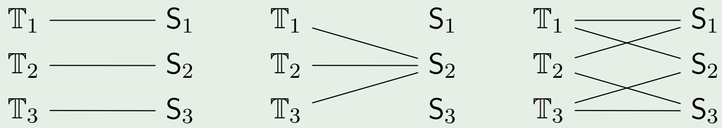


Universalism

Universalism

For all topics, there is at leas one inference system which is correctly applicable to it.

Diagrammatic representations

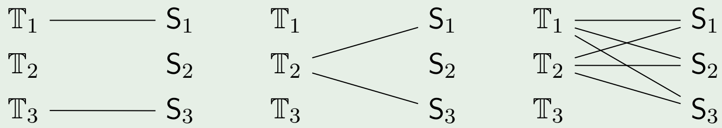


Anti-universalism

Anti-universalism

There is at least one topic to which no inference system is correctly applicable.

Diagrammatic representations

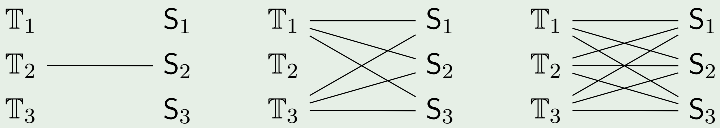


Topic-neutralism

Topic-neutralism

For any two topics (to which some inference system is correctly applicable), the inference systems correctly applicable to them are the same.

Diagrammatic representations

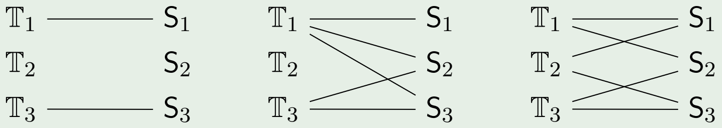


Topic-specificism

Topic-specificism

There are two topics (to which some inference system is correctly applicable) such that at least one inference system is correctly applicable to one but not to the other.

Diagrammatic representations



Topic-Sensitivity

Warning

Topic-specificness is not the same as topic-sensitivity (cf. Berto, *Topics of Thought*). In fact, the latter is compatible with topic-neutrality.

Topic-sensitivity

Topic-sensitivity is a property of some inference systems which make them distinguish whether two formulae may be about two different topics. Consider, e.g., the following sentences:

1. It is raining or it is not raining.
2. Laura is a musician or Laura is not a musician.

Given their distinct topics, however, these sentence are not logically equivalent in a topic-sensitive system.

In sum, *topic-sensitivity is a property of inference systems and not of logical views, like topic-neutrality.*

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The Question

The topics

Attempts to clarify what topics logic is allegedly topic-neutral about must be made against the background of the following question:

Do the laws of logic change, and, if so, how?

The contexts in which logic may be said to change are natural candidates for the notion of topic employed by the topic-specificist.

The Structure of Topics

Berto, *Topics of Thought*, p. 30

The space of topics, whatever these are, must display some mereological structure: topics can have proper parts; distinct topics may have common overlapping parts; one topic may be included in another in that every part of the former is also part of the latter. ... (Berto, *Topics of Thought*, p. 30)

Topics as Domains of Adequacy

P. Destouches-Février, *La structure des théories physiques*, 1951

D.-F. departs from Gonseth's view that logic has an empirical dimension: it is a 'physics' of the kinds of objects considered within a given domain. If scientific statements can all be represented as existential statements about abstract objects obeying the same laws, then classical logic would apply universally (pp. 4–5). Later, considering that different domains seem to have different physical laws, she argues that:

> [Since] a deductive theory is constructed to adequately describe a domain of science, and ... a logic must constitute the rules of reasoning of such a theory, then this logic is not independent of the content of this theory, of its domain of adequacy. The logic used to construct the theory must be adequate. There is no single logic that is independent of any content, but in each domain a logic is adequate. The logical and the physical, the formal and the real, are interdependent. (p. 88)

Topics as Domains of Adequacy

How does logic change in this view?

The laws of logic change across (scientific) domains, with each domain requiring the logical system that is adequate to its distinctive subject matter and laws of nature.

Critique: Haack, *Deviant Logic*, 1974

[W]hen two allegedly rival logical systems are said one to apply in one domain and the other in another, if the grounds for distinguishing the domains refers to content, one will be disposed to say that the systems are not really logical, and if the ground for distinguishing the domain refers to form, one will be disposed to say that the systems are not really rivals. (pp. 45–6)

[D.-F.] draws from her investigations into quantum logic the conclusion that there is no logic which works for any subject matter whatever. This, however, hardly follows: since all the principles of the weakest logic would presumably hold in any area of discourse. (p. 166)

Topics as Contexts

D. Batens, 'Meaning, acceptance and dialectics', 1985

> What I mean by 'a context' is precisely a communication situation. A context is characterized by (i) a set of participants, (ii) the problem that one tries to solve, (iii) the set of statements that are regarded as certain and in this sense define the set of possible answers to the problem, (iv) the set of aspects that are considered relevant to the problem, and (v) the set of methodological do's and don'ts that are judged appropriate with respect to the problem. (p. 334)

> [I]f some context proves inadequate to solve the problem under consideration, we have to move to another context to decide what is the right remedy; and in some cases the right remedy is to change the logic, and to change it only in some context, because only this context is affected by this inadequacy. (p. 358)

Topics as Contexts

How does logic change in this view?

The laws of logic change across (problem-solving) contexts, with a logical system being appropriate in case the inferential standards of a particular context proves inadequate for its aims.

Critique

- Context is not primarily defined by subject matter or domain, but by participants, problems, and methodological constraints.
- Differences in logic therefore track practical aims rather than differences in topics or domains.
- The variation in logic appears instrumental, since it depends on what is useful for solving a given problem.
- This suggests a pragmatic rather than substantive form of topic-specificism – perhaps compatible with the view of Beall and Restall (*Logical Pluralism*, 2006).

Topics as Regions in the Logical Space

O. Bueno, 'Modality and the plurality of logics', 2021

Things have properties and bear relations to other things . Some of these properties are modal: certain things are fragile, other are flammable. Some of these relations are modal: certain plants are edible by us, but not by other animals. The logical space records all of these possibilities. It is relative to these possibilities, conveyed in the logical space, that logical consequence relations, characteristic of the various logics, are constituted. (p. 324)

A logic is adequate [to a region] as long as the logical consequence relation it imposes on the various statements properly matches the relations among the various objects in [that region]. ...
Different logics sanction different possibilities: certain contradictory states ..., which are impossible according to classical logic, become possible in paraconsistent logics. Since the scope of the possible in the logical space is broader in paraconsistent logics, inferences that are taken to be valid in classical logic become invalid in a paraconsistent setting, such as explosion. As a result, on the view favored here, logic is not topic neutral: ... depending on the region of the logical space at stake, different logics are adequate. (p. 326)

Topics as Regions in the Logical Space

How does logic change in this view?

The laws of logic vary across regions of logical space, with different logics becoming appropriate depending on which possibilities are admitted in the region under consideration.

Critique

- If regions of logical space are logically distinguishable, a single logical system could capture these differences by introducing suitable logical terms related to these regions.
- In that case, the laws of logic would remain universal, there being no need for topic-specificness (cf. Zalta's object theory).
- Bueno also allows that more than one logic may be adequate for the same region of logical space.
- This seems to conflict with the requirement that a logic must properly match the relations among the objects in that region.

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The Received View About Logic

The Received View About Logic

Only classical logic (C) is correctly applicable to every topic.

Diagrammatic representations

$$\begin{tikzcd}[row sep=small] \mathbb{T}_1 \arrow[dr, dash] & \\ \vdots \arrow[r, dash] & \mathsf{C} \\ \mathbb{T}_n \arrow[ur, dash] & \end{tikzcd}$$

Remark

Even in the received view, there is a way in which the laws of logic may be said to change.

The Received View About Logic

How does logic change in this view?

The laws of logic vary as we focus on certain kinds of expressions.

Emergence of quasi-connexivity

Let the metavariables ‘$\mathtt{A}$’ and ‘$\mathtt{B}$’ below be placeholders for satisfiable formulae. Then, the following obtains (cf. SEP, ‘Connexive logic’, § 3.1):

1. $\mathtt{A} \not\models \neg\mathtt{A}$.
2. $\not\models \mathtt{A} \rightarrow \neg\mathtt{A}$.
3. $\mathtt{A} \models \mathtt{B} \Rightarrow \mathtt{A} \not\models \neg\mathtt{B}$.
4. $\models \mathtt{A} \rightarrow \mathtt{B} \Rightarrow \not\models \mathtt{A} \rightarrow \neg\mathtt{B}$.

Classical Connexivity

Strict conditional

Take the classical semantics of sentential logic and define a further, global operator '$\rightarrowtail$' as follows:

- $\hat{h}(\mathtt{A} \rightarrowtail \mathtt{B}) = \mathit{true} \Leftrightarrow$ for all ι, $\hat{\iota}(\mathtt{A} \rightarrow \mathtt{B}) = \mathit{true}$.

h, ι are two valued interpretations for atomic formulae and $\hat{h}, \hat{\iota}$ are their classical extensions to all formulae.

Emergence of connexive laws

Let, again, the metavariables '$\mathtt{A}$' and '$\mathtt{B}$' below be placeholders for satisfiable formulae. Thus, we have:

1. $\vDash \neg(\mathtt{A} \rightarrowtail \neg\mathtt{A})$.
2. $(\mathtt{A} \rightarrowtail \mathtt{B}) \vDash \neg(\mathtt{A} \rightarrowtail \neg\mathtt{B})$.

Classical Connexivity

W. Lenzen, 'A critical examination of the historical origins of connexive logic', 2020

Connexive principles in classical thinkers

The emergence of connexive laws within two-valued semantics is noteworthy in light of certain readings of the thinkers to whom connexive principles have been ascribed, namely Aristotle, Chrysippus, and Boethius.

Explanation of connexive principles

For instance, W. Lenzen suggests that, for these thinkers, the connexive principles were probably not meant to apply to all propositions, but just to 'normal' – that is, satisfiable – ones.

Against Topic-Specificness

Warning

The above does not mean that a two-valued or classical semantics would suffice for capturing all ways in which logical laws can change.

Remark

It nevertheless establishes that there is a way in which the laws of logic change without any appeal to specific topics – as we this change obtains within a view that is topic-neutralistic.

Hence, topic-specificness (and perhaps also pluralism) seems to be a superfluous view in logic.

On Non-Classicality

For monism

There may be a single logic able to capture these dynamics, but it needs not be the classical logic.

On non-classicality

Note, for instance, that classical logic obtains as a special case of some non-classical logics when we place restrictions on the expressions evaluated.

Thus, the one true logic – if there is one – may be non-classical.

Contents

Conclusions

- Logic has a topic – namely, correct reasoning, argumentation, or deduction – without thereby ceasing to be topic-neutral.
- Views about logic can be understood as assigning inference systems to the topics to which they correctly apply.
- Existing forms of topic-specificism use various conceptions of 'topic' – including scientific domains, problem-solving contexts, and regions of the logical space – fail to establish a genuine need for topic-specificism about logic.
- Changes in logical laws can already arise within a topic-neutralistic view by, e.g., restricting the class of expressions under consideration.
- Topic-specificness seems to be is theoretically superfluous.
- If there is a unique correct logic, it need not be classical.

Danke!

Funding

DAAD
Deutscher Akademischer Austauschdienst
German Academic Exchange Service